A

Python code defining the library

```python
import poolparty as pp

# Branch 0, a styled source pool + mutagenize
A = pp.from_seqs(['ACG', 'TGC'], prefix='a',
    style='green', mode='sequential')
B = A.mutagenize(num_mutations=1, prefix='mut',
                 style='orange', mode='sequential')

# Branch 1, a source pool
C = pp.from_seqs(['AAA', 'TGT', 'CTT'], prefix='c',
    mode='sequential')

# Merge branch 0 and 1 using the Stack Op
Final = pp.stack([B, C], prefix='stk')

df = Final.generate_library(num_cycles=1)
```

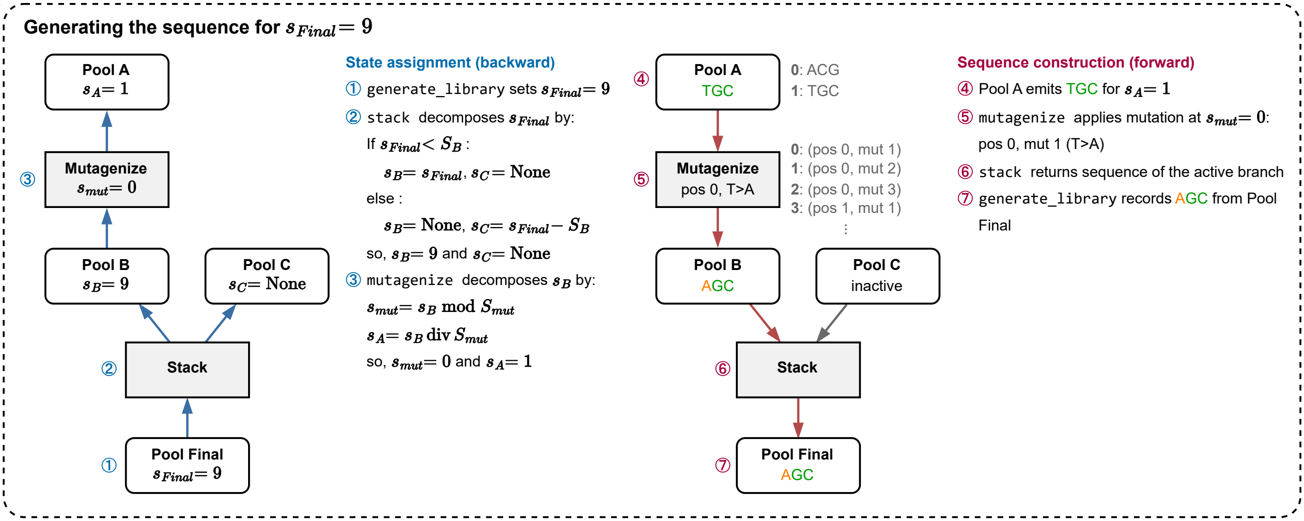
B

State counts

Pool / Operation	states	state rules
Pool A	$S_A = 2$	one state per sequence in from_seqs(...)
mutagenize	$S_{mut} = 9$	3 positions × 3 substitutions
Pool B	$S_B = 18$	$S_A \times S_{mut}$ = 2 × 9
Pool C	$S_C = 3$	one state per sequence in from_seqs(...)
stack	—	records active branch, not counted
Pool Final	$S_{Final} = 21$	$S_B + S_C$ = 18 + 3

Chaining an Operation multiplies the state count; stack adds it.
S = a number of states; s = one particular state.

C



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